NAGALAND LITERACY AND NUMERACY FEST 2026

Micro-Improvement Project (Science Teacher) Grade 6

Chapter 6 — Materials Around Us

Why Micro-Improvement Project (MIP)?

Across Nagaland, every object a student touches is made of something: a bamboo mat, a cane basket, a steel dao, a woollen shawl, a clay pot, or a plastic bottle bought at a Kohima or Dimapur market. This Micro-Improvement Project (MIP) helps students look closely at these everyday objects and sort them the way a scientist would by shine, hardness, whether they dissolve, whether they sink or float, and how much light passes through them rather than only by what they are called. It builds directly on students' existing sense of 'this is soft' or 'this shines' from daily life, and prepares them to plan and carry out simple investigations. Along the way, it also connects to a question every village and town in Nagaland faces today: why plastic litter stays in our rivers, like the Doyang and the Tizu, and our forests for years, while materials such as banana leaf, bamboo, and cloth do not and how understanding properties helps us choose better.

Skill in Focus	Materials Around Us — Sorting and Classifying by Properties
Duration	1 Week
Competency Addressed	C-1.3: “Asks questions and makes predictions about patterns.”

Learning Outcome

- Lists objects around us and analyses the materials they are made of.
- Plans and conducts a simple investigation to classify different materials by their observable properties.
- Observes the appearance of materials to differentiate them on the basis of lustre (shine).
- Examines materials by compressing or scratching them to categorise them as hard or soft.
- Observes what happens to materials added to water to categorise them as soluble or insoluble.
- Plans and conducts an investigation to find out whether various objects and materials sink or float in water.

- Classifies objects as opaque, transparent, or translucent.

Key Concepts

- Sorts and groups everyday objects by the material they are made of.
- Demonstrates sorting and grouping of materials according to common properties: appearance/lustre, hardness, solubility, density (floating and sinking), and transparency.
- Plans a simple, fair investigation to test and compare a property across several materials.
- Provides counter-examples to common misconceptions about materials (e.g. that bigger or heavier always means harder).

Detailed Tasks

Task-1	What Is It Made Of? — Objects Around Us

Objective	Build an intuitive starting point for the whole project — that every object around us is made of one or more materials — by listing and naming objects and their materials.
Setting	Outdoor (school ground/veranda) or Indoor (classroom)
Teacher's Note	This can be done as a short walk around the school ground/veranda pointing at real objects, or entirely indoors using classroom objects if outdoor space or time is limited. Keep the pace brisk and playful — this is a warm-up, not a test.
Duration	30 minutes
Materials Needed	One plastic object, one piece of wood, cane, or bamboo, one piece of cloth, and one metal object (spoon/dao/utensil) — whatever is already in the classroom or school · notebook and pencil

Activity Steps

1. Take students around the classroom, veranda, or school ground (or simply look around indoors) and ask: 'What is this made of?' for 5–6 objects — a bench, a window, a broom, a water bottle, a mat.

2. In pairs, students list 8–10 objects they can see and write down the material of each in a two-column table: Object | Material.
3. Ask: 'Which objects are made of more than one material?' (e.g. a pencil = wood + graphite + sometimes metal and rubber). Discuss 2–3 examples as a class.
4. Introduce the term formally: the substance used to make any object is called a material. Ask students to name the five materials they saw most often on their list.
5. Close with a discussion: 'If plastic did not exist, what would we use instead for each plastic object on your list?' Take 2–3 responses to set up the week's thinking.

Concept Built: *Every object is made of one or more materials; the same object can sometimes be made from more than one material.*

Task-2 | Sorting Shelf — Planning an Investigation to Classify Materials

Objective	Introduce the idea of planning a fair, simple investigation, and show that the same set of objects can be grouped in different ways depending on which property is chosen.
Setting	Indoor (Classroom)
Teacher's Note	This task sets up the method students will reuse for the rest of the week: choose one property, test every object the same way, then record and compare. Collect 8–10 varied classroom or household objects in advance (a stone, a coin, a leaf, a rubber band, a spoon, a plastic cap, a piece of chalk, a cotton ball).
Duration	30 minutes
Materials Needed	8–10 mixed small objects/materials already available in the classroom or brought from home · chart paper or the classroom floor marked into zones with chalk · notebook

Activity Steps

1. Show the class the 8–10 objects and ask: 'If I asked you to sort these into two groups, how many different ways could you do it?' Take 2–3 suggestions (by colour, by size, by what they are made of).
2. In groups, students choose one property (for example, 'hard vs soft', just by feel) and sort all the objects into two zones marked with chalk or on chart paper.
3. Ask a second group to re-sort the very same objects using a different property (for example, 'shiny vs dull'). Compare: did the groupings come out the same?
4. Discuss as a class: 'Since the objects did not change, why did our groups look different?' Introducing the idea that a material can be classified in many correct ways depending on which property we are testing, this sets up Tasks 3–6, where the class will test one property carefully at a time.
5. Explain the plan for the week on the board: day-by-day, the class will test lustre, hardness, solubility, sink/float, and transparency for the same list of objects, building one master properties table.

Concept Built: *The same set of materials can be classified in different, equally valid ways depending on which property is chosen; planning a fair test means changing only one property at a time.*

Task-3 Shine or Dull? — Lustre and Appearance

Objective	Help students observe and classify materials based on lustre — whether a surface shines or looks dull.
Setting	Indoor (Classroom)
Teacher's Note	Bring a genuinely shiny object (steel spoon, coin, mirror piece) and a genuinely dull one (chalk, cloth, mud) to demonstrate the contrast clearly before students test their own list.
Duration	30 minutes
Materials Needed	A steel spoon or coin · a piece of chalk, cloth, or wood · the class object list from Task-2 · notebook for the properties table

Activity Steps

1. Show the shiny steel spoon and the dull piece of chalk side by side. Ask: 'What is different about how light behaves on these two surfaces?'
2. Explain: objects that shine are called lustrous or shiny; objects that do not shine are called dull. This property is called lustre.
3. In groups, students test each object from their Task-2 list by holding it under light and observing whether it shines. They record 'Shiny' or 'Dull' in their properties table next to each object.
4. Discuss local examples: bamboo and cane objects are usually dull/natural in appearance, while polished steel utensils and new plastic items are often shiny. Ask: 'Does a shiny object always mean it is a better material?'
5. Group Task: students identify one object in their list where they were surprised by its lustre, and explain why to the class.

> **Concept Built:** *Lustre is the property of a material's surface to shine (be lustrous) or not shine (be dull), observed directly with the eyes.*

Task-4 Squeeze and Scratch — Hard or Soft

Objective	Help students classify materials as hard or soft by testing how easily they can be pressed or scratched, and correct the misconception that bigger or heavier objects are always harder.
Setting	Indoor (Classroom)
Teacher's Note	Supervise the scratch test closely — students should press or scratch gently with a fingernail or blunt pencil end, never with anything sharp, and never test anything that could break or injure.
Duration	30 minutes
Materials Needed	A stone or pebble · a cotton ball or piece of foam · a rubber band · a piece of wood · the class object list · notebook

Activity Steps

1. Demonstrate: try to press a stone with your finger (it does not give way) and then press a cotton ball (it presses easily). Explain: materials that are difficult to press or scratch are called hard; materials that press or scratch easily are called soft.
2. In groups, students gently test each object from their list — pressing or lightly scratching with a fingernail — and record 'Hard' or 'Soft' in their properties table.
3. Myth-busting moment: present two objects of very different size, e.g. a large cotton pillow and a small steel nail. Ask: 'Which is harder?' Students test and discover the small nail is harder despite being much smaller and lighter.
4. Discuss as a class: 'Does being bigger or heavier always mean something is harder?' Guide students to the counter-example above as proof that size and hardness are not the same property.
5. Differentiation Tip: for learners who find the distinction difficult, use very contrasting pairs first (stone vs cotton) before moving to closer comparisons (wood vs hard plastic).

Concept Built: *Hardness is tested by how easily a material can be pressed or scratched; a bigger or heavier object is not necessarily harder — demonstrated through a counter-example.*

Task-5	Sink, Float, or Dissolve? — Density and Solubility

Objective	Guide students to test and record, through a hands-on pattern table, which materials sink or float in water, and which dissolve or do not dissolve, then generalise a simple rule for each.
Setting	Outdoor (if using buckets/a tap) or Indoor (using mugs/bowls)
Teacher's Note	This is the most hands-on task of the week — plan for some water spillage. If outdoor tap or courtyard space is available, do this outside; otherwise, use bowls or mugs of water at each group's desk with a cloth ready for spills.
Duration	35 minutes
Materials Needed	A bucket, mug, or bowl of water per group · a stone, a dry leaf, a small piece of wood, a plastic cap, and a coin (for sink/float) · salt, sand, and sugar or chalk powder (for solubility) · notebook for the pattern table

Activity Steps

1. Sink or float: students place each object (stone, dry leaf, wood piece, plastic cap, coin) into the water one at a time and record 'Sinks' or 'Floats' in a table: Object, Material, Sinks/Floats.
2. Ask students to look for a pattern: 'Do all light objects float, and all heavy objects sink? Can you find an exception?' (A small dense coin sinks despite being small; a large dry log floats despite being big.) Guide them to the idea that it is not just size or weight, but how tightly packed the material is (its density) compared to water.
3. Soluble or insoluble: students add a spoon of salt to one mug of water and stir, then a spoon of sand to another, and observe. Record: does the material disappear and mix completely (soluble), or stay visible (insoluble)?
4. Repeat with sugar or chalk powder, and one more locally available substance the teacher chooses. Add both results to the same pattern table under a second column: Soluble/Insoluble.
5. Group Task: students predict, before testing, whether a piece of banana leaf and a small plastic bead will sink/float and dissolve/not dissolve then test to check their prediction, discussing any surprises.

Concept Built: *Materials can be classified as soluble or insoluble in water, and as sinking or floating, based on simple hands-on testing and pattern-based prediction rather than only size or weight.*

Task-6	See-Through Search

Objective	Apply the idea of transparency to real objects at home, classifying them as opaque, transparent, or translucent, and connect classroom learning to daily life.
Setting	Homework — done at home, discussed in class the next day
Teacher's Note	Explain this one day in advance with a clear example of each type (a wall is opaque, a clear glass is transparent, a thin curtain is translucent) so students know what to look for at home.
Duration	20 minutes
Materials Needed	Objects already at home — a window, a bottle, a curtain, a door, a plate — no purchase needed · notebook and pencil

Activity Steps

1. At home, students choose 5 objects (for example: a window pane, a water bottle, a wooden door, a thin curtain, a steel plate) and check how much they can see through each one.
2. Students classify each object as opaque (cannot see through at all), transparent (can see clearly through), or translucent (can see partly, but not clearly), and record this in a simple table: Object | Opaque/Transparent/Translucent.
3. Students ask one family member: 'Why do you think our windows are made of glass and not wood or cloth?' and note the answer in one sentence.
4. The next day, students share their five objects and classifications with a partner or the class, and add any new objects to the class's master properties table from the week.

Concept Built: *Transfers the classification of opaque, transparent, and translucent materials to real, self-chosen objects at home, strengthening the link between classroom science and daily life.*

Culminating Showcase

Showcase	**Nagaland Material Detectives — Choosing Better Alternatives**

Objective	Bring together every property tested in Tasks 1–6 into one creative, real-world investigation where groups compare a plastic object with a natural alternative and present their findings.
Setting	Indoor (combined classroom project) — 2–3 sessions after Task-6
Teacher's Note	Spread this across 2–3 sessions after Task-6, culminating in a short class exhibition. Encourage groups to bring real objects and alternatives from home wherever possible, and to be creative with their display.
Duration	30–35 minutes per session (2–3 sessions)

Materials Needed	Chart paper or cardboard · ruler, pencil, crayons or sketch pens · 5 plastic objects and their natural alternatives (steel bottle, cloth bag, banana-leaf plate, bamboo container, paper bag) — brought from home or made in class, wherever possible

Activity Steps

1. In groups of 4–5, students choose 5 plastic objects commonly used at home (for example: a plastic bottle, polythene bag, plastic plate, plastic basket, plastic pen) and one natural alternative for each.
2. Using the master properties table built across the week, groups test and record all five properties: lustre, hardness, solubility, sink/float, and transparency for both the plastic object and its alternative.
3. Groups decide, using their property comparisons, which material is better for the environment and for daily use, and note at least one property-based reason for each choice.
4. Groups identify and label at least one place in their design where they correct a misconception from Task-4 or Task-5 for instance, showing that a lighter alternative is not necessarily weaker or worse.
5. In the final session, each group presents its comparison chart in a short class exhibition, explaining their five objects, their property test results, and why their chosen alternatives are better for Nagaland's rivers, forests, and homes.

Concept Built: *Independent application of property-based classification, investigation, and misconception-awareness in one integrated, creative, environmentally-aware design task.*

BY THE END OF THIS PROJECT, TEACHERS ALONG WITH STUDENTS ARE REQUIRED TO COME UP WITH INNOVATIVE PROJECTS WHICH CAN BE EXHIBITED WITHIN THE CLASSROOM.

By the end of these tasks, teachers and students should have moved from simply naming objects to confidently sorting, testing, and comparing materials by their properties — culminating in an original, locally-grounded comparison of plastic and its alternatives that each group can showcase with pride.

Reflect on Students' Understanding

Observe 5–6 students each week during the tasks above. No written test is needed — watch what students do with real objects and water, and listen to how they explain their thinking.

Student Name	Classifies by Property	Tests Hardness	Tests Sink/Float & Solubility	Identifies Transparency	Needs Support
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐

Scale (for each column): ☐ Not Yet ☐ Sometimes ☐ Consistently

Picture Guide